TITLE

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Abstract

Keywords: Multi-agent systems, Vision-language models, Double reading, Visual grounding, Error correlation

1. Material and methods

1.1. Study design

This reader study evaluated single readers and reading panels of artificial intelligence (AI) vision-language models answering closed questions about radiology images. Controlled manipulations varied the information available to readers and the components they shared. The design distinguished image evidence from the question-text prior, defined as information supporting an answer from the question and options without image content. It also separated seeing, the extraction of visual findings, from judging, the selection of an answer using findings and prior knowledge. The protocol and validation decisions were locked before the full runs. Pilot observations were excluded from the study analyses.

1.2. Data sets and sampling

Five item sets were assembled from public data sets. S1 contained 250 of 251 eligible VQA-RAD test questions with yes/no answers.[1] Duplicate image–question combinations were removed before sampling. S2 contained 250 of 355 eligible English SLAKE test questions labeled as closed and answered yes/no.[2] S3 contained 250 of 840 eligible MedXpertQA-MM items for which every attached image was a radiologic acquisition.[3] Image modality was labeled by gpt-4.1-nano, and claude-haiku-4.5 confirmed that candidate images were actual acquisitions. Drawings, diagrams, photographs, pathology slides, charts, and tables were excluded.

S4, SLAKE-target, contained 250 of 335 eligible English closed yes/no questions from the SLAKE training and test sets, excluding S2 items. Questions concerned an organ or lesion present in the image and localized by the supplied annotations. The expanded target bounding box occupied at most 40% of the image. Each item also had an equal-sized, nonoverlapping control box with at least 70% of its area inside the body. S5, SLAKE-pairs, contained 200 pairs, or 400 questions, spanning 104 question templates. Each pair used the same question with different images and opposite reference answers. Sampling allowed at most 5 pairs per template.

Sampling used stratified shuffling with seed 42. Strata were reference answer for S1, reference answer and modality for S2, body system and medical task for S3, and target type and modality for S4. S5 was stratified by question template. Yes/no option order was randomized separately for each item using its identifier. Images were resized proportionally, without upsampling, to at most 1024 pixels on the long side.

1.3. Models

The application programming interface (API) models were gpt-5-mini, gemini-3.7-flash, claude-haiku-4.5, and qwen3.8-flash. All reader calls used OpenRouter; the routing and modality-screening outputs of gpt-4.1-nano (temperature 0) were generated once and reused. Where a model accepted the parameter, panel readers, discussion rounds, and report writers sampled at temperature 0.7 and the single reader at 0.3; the seed was 1, and gpt-5-mini used low reasoning effort. Google models were restricted to Gemini Flash. Routing and token budgets were held constant within each experimental condition. Because of its throughput, qwen3.8-flash ran only the original-image baseline (A0), graded evidence (A1), generic reading panel (C1), and varied image rendering (C2).

Five local open-weight models were evaluated: MedGemma-4B, Qwen3-VL-8B, Qwen2-VL-7B, HuatuoGPT-Vision-7B, and LLaVA-1.5-7B.[4–8] HuatuoGPT-Vision and LLaVA-1.5 shared the Contrastive Language–Image Pretraining (CLIP) ViT-L/14-336 vision encoder.[9] LLaVA-Med v1.5 was screened and excluded under the prespecified accuracy rule: its accuracy was 53%, below the 55% threshold, pooled over 100 validation items. These comprised 50 items each from S1 and S2. LLaVA-1.5 met the pooled threshold; its VQA-RAD validation accuracy was near chance.

1.4. Readers and reading panels

The single reader received a general-radiologist role and instructions to reason briefly, step by step, before selecting an answer. For each specialty reading panel, gpt-4.1-nano ranked 9 radiology subspecialty roles using the question text. The roster covered neuroradiology, head and neck, thoracic, cardiovascular, abdominal, genitourinary, musculoskeletal, breast, pediatric, nuclear medicine, interventional, and emergency radiology. Panels of 1, 3, and 9 readers were nested: smaller panels used the first members of the same ranked roster and reused their initial answers. Answers were aggregated by majority vote.

Discussion panels started from these independent answers and allowed up to 3 subsequent rounds, stopping at unanimity. Readers reviewed peer answers and reasoning and could retain or revise their answers. Peer context was capped at 900 characters, with peer order shuffled by item and round. Generic panels contained 9 readers with identical general-radiologist prompts; their initial responses differed only through separate sampling.

1.5. Experiment A: image evidence and question-text prior

Experiment A used S1–S5. The original-image baseline compared the single reader, independent specialty panels, and discussion panels of 3 and 9 readers. Graded evidence compared original images capped at 1024 pixels, images capped at 256 or 128 pixels, and uniform gray images. Gray images retained the original rendered dimensions, and prompts and options remained unchanged. The single reader and independent specialty panels were evaluated at each evidence level.

On S4, the annotated target bounding box was expanded by 10% on each side and clipped to image boundaries. Target and control boxes were filled with mid-gray before resizing. Control boxes matched target-box dimensions, did not overlap the target, and met the body-coverage criterion. On S5, readers answered each member of the same-question, different-image pairs separately using the original images.

1.6. Experiment B: seeing and judging

Experiment B used S1–S3 and crossed 3 perceiver models with 3 reasoner models: gpt-5-mini, gemini-3.7-flash, and claude-haiku-4.5. Each perceiver received images without questions or answer options. It wrote findings reports of at most 120 words describing modality and plane, anatomy shown,

and every abnormal or notable finding with its location. Reports were reused across questions about the same image.

Reasoners received the report, question, and options without the image. In the independent-report condition, each of 9 readers received its own report; all perceiver–reasoner combinations were evaluated. In the shared-report condition, all readers received the first report; this condition used matching perceiver and reasoner models. Generic panels reading images directly provided the reference comparison. For S2, report coverage of the queried organ or lesion was assessed using a synonym list fixed from SLAKE labels before running this pathway.

1.7. Experiment C: shared components

Experiment C used S1–S3 to vary shared vision-language models, image rendering, roles, reports, reference cases, and discussion records. In the rendering condition, generic readers 1–3 viewed original images, readers 4–6 viewed gamma-transformed images with gamma 0.7, and readers 7–9 viewed contrast-limited adaptive histogram equalization (CLAHE) images.[10] CLAHE used a clip limit of 2.0 and an 8×8 tile grid on the lightness channel. Gamma 0.7 replaced grayscale inversion because inversion reduced mean single-reader accuracy by 6.6 percentage points in validation, exceeding the prespecified 5-percentage-point limit. Cross-vendor comparisons combined existing reader responses without additional calls.

Reference-case memory was evaluated on S1 and S2 using VQA-RAD and English SLAKE training images with closed yes/no questions. Images identical to any test image were excluded by SHA-256 hash. For each query, exclusions also removed references within Hamming distance 8 of its 64-bit perceptual hash or with CLIP cosine similarity of at least 0.98. The data sets carried no patient identifiers; these bounds guarded against references from the same scan or patient.

Retrieval used CLIP ViT-L/14-336 image embeddings. The 27 nearest eligible images were shuffled deterministically by question identifier into 9 sets of 3. Shared memory supplied set 1 to every reader; private memory supplied set i to reader i. Each reference included its image and a closed question–answer pair. The question was selected by maximal term frequency–inverse document frequency (TF-IDF) cosine similarity to the current question. Text-only memory supplied the same question and answer without the reference image. Single readers also received image-based or text-only references.

1.8. Experiment D: open-weight models

Experiment D used S1–S4 and the same rendered pixels as the API arm. Conditions included original and gray images, 256- and 128-pixel images, gamma 0.7, CLAHE, and target and control occlusion on S4. Each item and condition required one forward pass without sampling. Option probabilities were obtained by applying softmax to next-token logits for the answer words yes/no on binary items and option letters otherwise. The highest logit among accepted capitalization and leading-space token variants represented each answer. Letter readout for yes/no items was rejected during validation because the LLaVA family assigned 96%–100% probability to "A" regardless of image content.

1.9. Outcome measures

Accuracy was the proportion of answers matching the reference. Error correlation, ϕ, was the mean Pearson correlation of binary error indicators across items over reader pairs. Pairs with constant error indicators were omitted. The effective number of independent readers was $N_{\text{eff}} = 9/(1+8\phi)$. For readers of unequal accuracy, correlation was also divided by its maximum given their marginal error rates, $\phi_{\max}$.

The prior decomposition cross-tabulated each reader's correctness with real and gray images. Consensus errors required at least 7 of 9 readers to be wrong; their majority answers were compared with gray-image majorities. Flip rate measured the proportion of answers changed by occlusion. Region dependence was target flip rate minus control flip rate, calculated for readers and panel majorities. Pair accuracy required both paired answers to be correct; the proportion of pairs answered differently was also measured. Report-pathway analyses compared accuracy and error correlation and separated errors by report coverage of the queried target.

The independence-budget analysis regressed pairwise error correlation on indicators of different perceivers, reasoners, renderings, roles, reference-case sets, and reports. Ordinary least squares estimated changes in ϕ; discussion was evaluated separately by comparing initial and final correlations. Conformity was the proportion of initially correct readers who ended wrong, stratified by whether their initial answer changed between real and gray images.

For open-weight models, visual reliance was the absolute change in log-odds of yes between real and gray images. For multiple-option items, reliance

was the Kullback–Leibler divergence from the real-image to gray-image probability distribution. Region reliance subtracted the corresponding control-occlusion change from the target-occlusion change. Model-pair error correlations were compared by shared vision encoder, and Spearman correlation related API consensus errors to open-weight visual reliance. Capture rate was $C = (A_{\text{panel}} - A_{\text{single}})/(A_{\text{oracle}} - A_{\text{single}})$, where A denoted accuracy. The oracle accepted any correct initial reader answer. The symbol C distinguished this measure from Cohen's or Fleiss' kappa.

1.10. Statistical analysis

Percentile bootstrap 95% confidence intervals used 2,000 item-level resamples with seed 0, including paired differences. Error-correlation intervals used 500 resamples; independence-budget regression and its discussion comparisons used 300 resamples. Resampling retained all reader responses and matched conditions for each item; paired-image measures resampled complete pairs. Unparsed answers counted as wrong. Analyses used descriptive estimates and confidence intervals without hypothesis tests or P values.

1.11. Validation before full runs

V1 checked API answer parsing and reasoning effort; V2 checked image transformations; V3 checked reference retrieval, leakage exclusions, and multiple-image parsing. V4 checked target and control boxes and image pairs; V5 checked report length, completeness, and answer parsing; V6 checked local-model probabilities and accuracy eligibility. Full runs used only the settings fixed by these checks.

1.12. Code and data availability

The code is public at `https://github.com/hq0709/consultation-arithmetic`. VQA-RAD, SLAKE, and MedXpertQA are public data sets.

Authors' contribution

TBD.

Ethics statement

The study used only public, de-identified data sets and involved no human participants. Institutional review board approval was not required.

Conflicts of interest

The authors declare no conflicts of interest.

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